

CSA0614

## DESIGN AND ANALYSIS OF ALGORITHMS

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### Experiment 18: Selection Sort

#### Code:

```
def reorder_shelf(arr):
    arr = arr.copy()
    moves = 0
    n = len(arr)
    for i in range(n):
        min_index = i
        for j in range(i + 1, n):
            if arr[j] < arr[min_index]:
                min_index = j
        if min_index != i:
            arr[i], arr[min_index] = arr[min_index], arr[i]
            moves += 1
    return arr, moves

books = [305, 102, 250, 118, 199, 400, 101]
ordered, moves = reorder_shelf(books)
print("Ordered Books:", ordered)
print("Physical Moves:", moves)
```

#### Output:

```
=== RESTART: C:/Users/hk838/Downloads/Slot- A,B,C,D/
Ordered Books: [101, 102, 118, 199, 250, 305, 400]
Physical Moves: 4
```